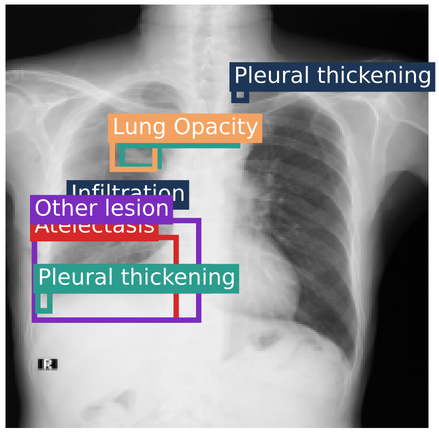
# Presentation Comparison Report

Hard diffuse case with partial model capture    Case: 305e4add9c72c91e9984305bf4e85aee

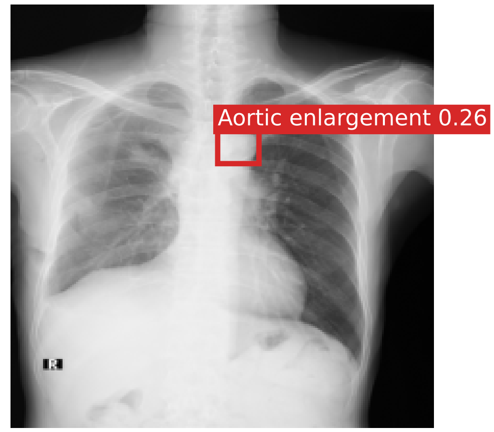
Review status: needs_human_review    Confidence: moderate

Original CXR

Ground-truth boxes




YOLO support overlay




## Ground Truth

- Atelectasis
- Infiltration
- Lung Opacity
- Other lesion
- Pleural effusion
- Pleural thickening
- Pneumothorax
- Pulmonary fibrosis

## Swin Classification

Predicted: Lung Opacity, Pleural effusion, Pleural thickening, Pulmonary fibrosis
Correct: Lung Opacity, Pleural effusion, Pleural thickening, Pulmonary fibrosis
Missed: Atelectasis, Infiltration, Other lesion, Pneumothorax
Overcalled: None

## YOLO Labels

Predicted: Aortic enlargement
Correct: None
Missed: Atelectasis, Infiltration, Lung Opacity, Other lesion, Pleural effusion, Pleural thickening, Pneumothorax, Pulmonary fibrosis
Overcalled: Aortic enlargement

## Agentic Workflow Summary

Supported findings: Pulmonary fibrosis, Pleural thickening, Pleural effusion, Lung Opacity
Uncertain findings: Aortic enlargement
Final report labels: Lung Opacity, Pleural effusion, Pleural thickening, Pulmonary fibrosis
Pattern buckets: Pleural abnormality, Airspace or infectious-inflammatory pattern, Chronic interstitial or fibrotic pattern, Possible urgent thoracic abnormality

# Evidence and Report Summary

## Findings

The chest X-ray demonstrates findings consistent with chronic pulmonary changes including pulmonary fibrosis and pleural abnormalities. There is evidence of pleural thickening and pleural effusion. Lung opacity is present, which may represent chronic interstitial changes or acute airspace disease. A possible aortic enlargement is detected but requires confirmation given conflicting model evidence.

## Impression

Findings suggest chronic interstitial lung disease with pulmonary fibrosis and pleural abnormalities including thickening and effusion. The presence of lung opacity may indicate either chronic fibrotic changes or superimposed acute process. Possible aortic enlargement requires clinical correlation. These findings warrant urgent clinical evaluation given the potential for acute components.

## Top Swin Probabilities

- Pulmonary fibrosis: 0.981
- Pleural thickening: 0.969
- Pleural effusion: 0.944
- Lung Opacity: 0.695
- Aortic enlargement: 0.402
- Infiltration: 0.371

## Ground-Truth Boxes

- Atelectasis @ [34.2, 281.1, 206.0, 381.2]
- Infiltration @ [78.4, 242.7, 139.1, 275.4]
- Infiltration @ [140.4, 168.4, 185.1, 194.8]
- Lung Opacity @ [128.7, 162.0, 180.2, 198.6]
- Other lesion @ [34.2, 260.6, 233.3, 381.2]
- Pleural effusion @ [38.2, 344.2, 52.9, 370.0]
- Pleural thickening @ [276.0, 99.9, 291.3, 115.2]
- Pleural thickening @ [38.2, 344.2, 52.9, 370.0]

## Top YOLO Detections

- Aortic enlargement: 0.26 @ [249.9, 151.3, 299.9, 191.7]

## Thresholds Used

Swin positive label cutoff: 0.50
YOLO detection confidence cutoff: 0.25
Agentic AI confidence band is qualitative, not a fixed numeric threshold.

## Presentation Takeaway

Swin correct labels: 4 / 8
YOLO correct labels: 0 / 8
Key miss for Swin: Atelectasis, Infiltration, Other lesion
Key miss for YOLO: Atelectasis, Infiltration, Lung Opacity

# Confidence Calculation

## How this case's confidence was calculated

Per-finding band cutoffs: high >= 0.80, moderate >= 0.50, low < 0.50.
Swin label threshold = 0.50. YOLO detection threshold = 0.25.
Pulmonary fibrosis: high from score = Swin(0.98)
Pleural thickening: high from score = Swin(0.97)
Pleural effusion: high from score = Swin(0.94)
Lung Opacity: moderate from score = Swin(0.70)
Aortic enlargement: low from score = max(Swin(0.40), min(1.00, 0.6*0.40 + 0.4*0.26)) = 0.40
Agentic AI report band reference: `moderate` occurred in 58 / 300 cases with exact-match accuracy 15.5%.

Interpretation note: the report-level agentic AI confidence band is constrained to high / moderate / low in the schema, but the model chooses that band. The per-finding confidence lines above are deterministic and computed from the Swin probability, YOLO support, and the label taxonomy.